

Summary: Matsushita (2015), "Japanese Forestation Policies During the 20 Years Following World War II"

Source: Koji Matsushita, Chapter 4 in *Precious Forests - Precious Earth*, IntechOpen, 2015. DOI 10.5772/61268. Open access.

Core argument

Detailed policy history of how Japan solved its post-WWII "cutover land crisis" (excess wartime cutting, no replanting) through three successive national forestation plans (1949-1960), then pivoted to a coniferous-expansion policy that intensively planted sugi/hinoki across the country — the single biggest cause of the age-class imbalance and forestry-corporation debt crisis Japan lives with today. This is the direct policy-level companion to the demographic/ecological picture in [[project_reforestation_principle]]-adjacent sources (Iwamoto, Yamaura, Leiter).

The wartime/postwar crisis (1929-1949)

- From 1929-1949, annual cut area nationally exceeded planted area every year; the gap peaked in the mid-1940s. Total cumulative excess-cut area 1922-1955 was **2.79 million ha** (1.44M ha national forest + 1.36M ha non-national).
- By end of 1948, cutover land totaled **1.49 million ha** (≈6% of Japan's total forest area) — concentrated in the most accessible forests near villages.
- Cut volume of industrial roundwood exceeded coniferous growth throughout the 1940s-50s; multiple contemporary studies (GHQ, Japanese Forestry Agency, Nihon Ringyo Gijyutsu Kyokai) forecast domestic wood exhaustion within 15-33 years depending on assumptions used — creating strong political urgency for the plans that followed. (Matsushita later shows these estimates were methodologically shaky — Japan never actually ran out.)
- Reasons cited for slow postwar replanting: economic instability, high planting costs, land-reform anxiety, labor/material/food/money shortages, and forest tax (nine factors reported by Nihon Ringyo Gijyutsu Kyokai).

Three sequential national forestation plans

1. **5-year Forestation Plan (1949-1953):** first full-scale postwar plan, targeting the 1.5M ha of cutover land; 2.82M ha total planned. Budget shortfalls limited actual execution to 60-70% of plan.
2. **10-year Forestation Plan (1952-1961):** introduced under the 1951 amended Forest Act, which for the first time made planting *compulsory* — prefectural governors could designate sites and legally obligate owners to plant (Article 14), a genuinely exceptional intervention in Japan's otherwise non-compulsory forestry system. Goal: industrial-roundwood self-sufficiency via coniferous expansion (0.84M ha of broadleaf→conifer conversion planned). Achieved its highest-ever annual planted area in 1954.
3. **6-year Forestation Plan (1955-1960):** shifted emphasis from *quantity* (war-damage reconstruction) to

quality — high-grade nursery stock, seed-tree systems, "right tree, right site" principle. Non-national planted forest area targeted to grow from 4.4M to 6.0M ha.

4. **1966 Basic Plan on Forest Resources** (first Cabinet-level plan under the 1964 Forestry Basic Act): the most ambitious — targeted **13.4 million ha** of planted coniferous forest by 2015 (actual outcome: ~10M ha) and planned annual cutting volume to reach 133M m³ by 2015 (actual 2012 figure: only 36.5M m³ — less than a third of plan). Matsushita identifies this over-optimistic 50-year demand assumption as the root planning failure behind today's oversupply and stumpage-price collapse.

Profit-sharing forestation and the corporation debt crisis

- To mobilize planting capital, prewar and postwar law (esp. the 1958 Act on Special Measures Concerning Profit-Sharing Forestation) let landowners, government, and private capital (notably pulp/paper and electric-power companies, who had regulatory planting obligations) split future harvest revenue.
- **Prefectural Forestry Corporations** (one per prefecture, founded 1959-1971) and the national **Forest Development Corporation** (1956) became the main vehicles, targeting the least economically attractive land — remote, poor-access, broadleaf-heavy, sparsely-owned areas — where private planting would never otherwise have happened. Combined they planted roughly 860,000 ha.
- **Fatal design flaw:** these Corporations funded *all* costs (planting, tending, thinning, roads, wages, interest) via long-term debt, expecting the eventual harvest revenue (40-50 years later) to cover it — under the assumption log prices would stay flat or rise, per the 1966 Basic Plan.
- Stumpage prices for sugi/hinoki instead collapsed after 1980 (price index of 100 in 1980 fell to ~11-15 by 2013). Corporations extended final-cutting ages to avoid harvesting at a loss, which only deepened the debt.
- Result: **all 47 Prefectural Forestry Corporations are now in serious financial distress**, with combined long-term debt of roughly **US\$8.7 billion** (¥1.04 trillion). Interest alone reached >90% of one sampled corporation's annual costs by the 2010s (46.4% of its total 1966-2012 costs). Several have since dissolved or been absorbed into prefectural forest systems.

Self-sufficiency outcome — the policy's central irony

- Despite achieving the coniferous-expansion goal (10M ha planted, ~40% of forest area, 3.0 billion m³ growing stock), Japan's **wood self-sufficiency rate fell to 27.9% by 2012** (bottomed at 18.2% in 2002) — the forestation policy succeeded at growing trees but trade/tariff liberalization (log tariffs dropped near-zero in 1951; foreign-exchange controls on log imports abolished 1960) meant domestic timber production was never actually protected. Matsushita calls the forestation policy and the trade policy of the same era "somewhat contradictory."
- By 2012, total annual growth (74M m³) slightly exceeded annual demand (72M m³) — Japan's forest volume has fully recovered in aggregate — but harvesting activity remains chronically weak, leaving the now-mature postwar cohort largely unharvested (echoes Leiter et al.'s 53-58% utilization finding and Yamaura et al.'s "reluctant to harvest" observation).

Author's conclusions

The cutover-land crisis and the domestic-wood-dearth crisis were both genuinely solved by these policies — but the government never planned an endpoint or an exit strategy for the coniferous-expansion program, and consequently: (1) forests are badly age-skewed and increasingly need thinning that isn't happening (aging owners/workers, no roads); (2) Prefectural Forestry Corporations carry unsustainable debt from a financing

model that assumed permanently rising log prices; (3) domestic self-sufficiency was undermined by simultaneous trade liberalization. Matsushita frames the urgent open question as: how should Japan now manage ~10 million ha of existing planted forest, in a new way, without repeating the same top-down single-scenario planning error.

Relevance to Mitsue project

This is the missing policy-history piece behind [[reference_fit_grid_biomass]], [[project_council_minutes_forestry_findings]], and the Leiter/Yamaura summaries already in this folder — it explains *why* Japan (and by extension Mitsue's own Sugi-heavy, aging, undermanaged stands) ended up here: a single-scenario, debt-financed, top-down national planting program with no exit plan, undermined by its own trade policy. Two points worth carrying into outreach material:

- The **Prefectural Forestry Corporation debt crisis** (~\$8.7B nationally) is a strong cautionary precedent for any financing structure proposed for Mitsue's own CHP/reforestation plan — avoid a model that assumes flat-or-rising future timber prices to justify debt-financed planting, and avoid open-ended commitments with no harvest/exit trigger.
- The 1966 Basic Plan's failure (targeted 133M m³/yr harvest by 2015, actual 36.5M m³) is a concrete national-level illustration of exactly the "under-harvested, over-planted" thesis that justifies positioning Mitsue's biomass CHP + DC as *demand* for wood Japan already has too much of, not a new extraction pressure.